

Measuring g from a projectile on a tilted air table

PHYS 1552 · Experiment 1 · Data analysis exercise: Investigation II and Analysis III

Up-slope velocity v_y vs time for three projectile launches (table tilt $\theta = 3.5^\circ$)

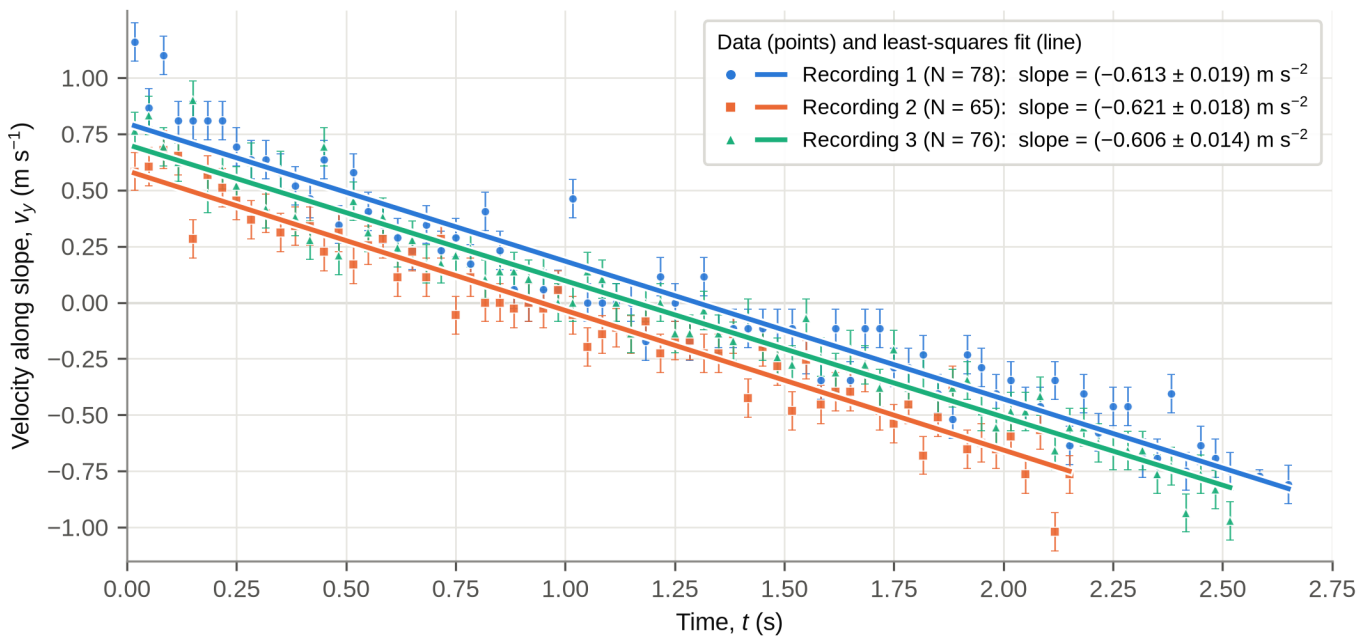


Figure 1. Velocity component along the slope, v_y (positive = up the slope), against time for three hand-launched projectile recordings on the air table tilted at $\theta = (3.5 \pm 0.1)^\circ$. Each point is $v_{y,av} = \Delta y / \Delta t$ between consecutive video frames ($\Delta t = 1/30$ s), plotted at the interval mid-point time. Error bars: ± 0.085 m s $^{-1}$, from a ± 2 mm uncertainty in locating the puck centre in Tracker in each frame, $\delta v_y = \sqrt{2} \cdot \delta y / \Delta t$; the timing uncertainty is negligible, so no horizontal bars are drawn. Lines: unweighted least-squares fits to all points of each recording; slope uncertainties are the standard errors of the fit (Excel LINEST).

What the graph shows. In all three recordings v_y falls linearly with time: positive while the puck moves up the slope, zero at the top of the parabola ($t \approx 0.9$ – 1.3 s) and negative as it comes back down. A straight v – t line means a constant acceleration equal to its slope. On the air cushion the only force along the table is the component of gravity down the slope, so $a_y = -g \sin \theta$ and $g = |\text{slope}| / \sin \theta$. Meanwhile v_x stays constant (group's averaged v_x fit: slope ≈ -0.01 m s $^{-2}$), so the horizontal motion is unaffected by the vertical acceleration — the two components are independent.

Slopes and their differences. Recording 1: (-0.613 ± 0.019) m s $^{-2}$; recording 2: (-0.621 ± 0.018) m s $^{-2}$; recording 3: (-0.606 ± 0.014) m s $^{-2}$. The largest difference (recordings 2 and 3, 0.015 m s $^{-2}$) is only $0.65\times$ their combined fit uncertainty, so the three slopes agree and the lines are parallel. They differ only in intercept, i.e. the launch velocity (0.80 , 0.59 and 0.70 m s $^{-1}$): each launch was different, but the acceleration does not depend on how the puck was launched.

Error bars and fit quality. The ± 0.085 m s $^{-1}$ bars are about 10–15 % of the launch speeds. The fitted lines pass through 45 %, 71 % and 66 % of the error bars (60 % overall), close to the ≈ 68 % expected for 1σ bars, so the ± 2 mm estimate is realistic. Recording 1 scatters more (residual SD 0.13 vs 0.09 m s $^{-1}$): its video scale is coarser (≈ 1.9 mm per pixel vs ≈ 1 mm) and its first launch points lie 0.1 – 0.4 m s $^{-1}$ above the line, so its bars are somewhat underestimated. Excluding the first 0.1 s of every recording changes the slopes by ≤ 0.03 m s $^{-2}$ and the final g to 9.80 m s $^{-2}$, within its uncertainty, so the conclusion does not depend on this choice.

g from each recording. $g_i = |\text{slope}_i| / \sin \theta$, with $\delta g/g = \sqrt{[(\delta a/a)^2 + (\delta \theta / \tan \theta)^2]}$ and $\delta \theta = 0.1^\circ$ (digital-level resolution), which alone contributes 2.9 %: $g_1 = (10.05 \pm 0.43)$ m s $^{-2}$, $g_2 = (10.18 \pm 0.42)$ m s $^{-2}$, $g_3 = (9.93 \pm 0.37)$ m s $^{-2}$.

Combined value and choice of uncertainty. I average the three slopes, $a = (-0.614 \pm 0.010)$ m s $^{-2}$, then convert to g . The ± 0.010 is propagated from the fit errors, $\delta a = \sqrt{(\sum \delta a_i^2)/3}$; the standard error of the mean of the three slopes is smaller, ± 0.004 m s $^{-2}$. I report the propagated value: the slopes agree within their fit errors, so the small SEM only reflects three values happening to land close together, and an SEM from just three values is itself very uncertain. The tilt error is shared by all recordings (one angle measurement), so it is systematic, does not average down, and is added once after averaging. Result: **$g = (10.05 \pm 0.33)$ m s $^{-2} \approx (10.1 \pm 0.3)$ m s $^{-2}$.** The angle term (2.9 %) dominates the slope term (1.6 %), so using the SEM instead would barely change it (± 0.30 m s $^{-2}$).

Comparison with the tabulated value. Our result is 0.24 m s $^{-2}$ (2.5 %) above $g = 9.81$ m s $^{-2}$, i.e. 0.7σ , so the two agree within one standard uncertainty; each individual g_i also agrees within 1σ . The small excess would be fully explained if the true tilt were 3.59° instead of 3.50° (within the level's resolution) or by a ≈ 2.5 % error in the Tracker length calibration. As the angle now dominates the uncertainty, measuring θ more precisely (e.g. from the height difference along the table) would improve the result most.

Raw data used for Figure 1

Screenshot of the group spreadsheet (Sheet1): Tracker output t , x , y and the computed v_x , v_y for recordings (trials) 1–3. Top: rows 1–54; bottom: rows 54–85. Figure 1 plots the v_y columns (E, L, S) against the mid-point time of each frame interval.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	
1					Theoretical g																			
2					-0.59828																			
3																								
4																								
5	Trial 1																							
6	t	x	y	vx (avr)	vy (avr)	t (avg)		Trial 2	t	x	y	vx(avg)	vy(avg)	t(avg)		Trial 3	t	x	y	vx	vy	t(avg)	Averages All	
7	0.00E+00	1.93E-03	3.85E-03	DNE	DNE	3.33E-02		0.00E+00	2.22E-03	8.89E-04				3.33E-02		0.00E+00	2.52E-03	1.35E-03					8.34E-01	2.24E-01
8	3.33E-02	9.63E-03	4.24E-02	2.31E-01	1.16E+00			3.33E-02	9.99E-03	2.03E-02	2.33E-01	5.83E-01				3.33E-02	9.45E-03	2.68E-02	2.08E-01	7.63E-01		7.68E-01	1.94E-01	
9	6.67E-02	1.35E-02	7.13E-02	1.16E-01	8.67E-01			6.67E-02	1.86E-02	4.05E-02	2.59E-01	6.05E-01				6.67E-02	1.64E-02	5.45E-02	2.08E-01	8.32E-01		8.24E-01	2.43E-01	
10	1.00E-01	1.93E-02	1.08E-01	1.73E-01	1.10E+00			1.00E-01	2.90E-02	6.32E-02	3.12E-01	6.80E-01				1.00E-01	2.45E-02	7.76E-02	2.43E-01	6.93E-01		6.95E-01	2.46E-01	
11	1.33E-01	1.93E-02	1.35E-01	0.00E+00	8.09E-01			1.33E-01	4.32E-02	8.49E-02	4.25E-01	6.52E-01				1.33E-01	3.49E-02	9.84E-02	3.12E-01	6.24E-01		6.65E-01	1.38E-01	
12	1.67E-01	2.31E-02	1.62E-01	1.16E-01	8.09E-01			1.67E-01	4.51E-02	9.44E-02	5.67E-02	2.83E-01				1.67E-01	4.30E-02	1.28E-01	2.43E-01	9.01E-01		6.20E-01	2.07E-01	
13	2.00E-01	2.70E-02	1.89E-01	1.16E-01	8.09E-01			2.00E-01	5.26E-02	1.13E-01	2.27E-01	5.67E-01				2.00E-01	5.22E-02	1.45E-01	2.77E-01	4.85E-01		6.36E-01	1.41E-01	
14	2.33E-01	2.70E-02	2.16E-01	0.00E+00	8.09E-01			2.33E-01	6.21E-02	1.30E-01	2.83E-01	5.10E-01				2.33E-01	5.68E-02	1.64E-01	1.39E-01	5.89E-01		5.56E-01	1.79E-01	
15	2.67E-01	3.08E-02	2.39E-01	1.16E-01	6.93E-01			2.67E-01	7.15E-02	1.45E-01	2.83E-01	4.54E-01				2.67E-01	6.14E-02	1.82E-01	1.39E-01	5.20E-01		5.43E-01	1.76E-01	
16	3.00E-01	3.66E-02	2.60E-01	1.73E-01	6.36E-01			3.00E-01	7.53E-02	1.58E-01	1.13E-01	3.68E-01				3.00E-01	6.95E-02	2.02E-01	2.43E-01	6.24E-01		4.83E-01	1.34E-01	
17	3.33E-01	3.85E-02	2.81E-01	5.78E-02	6.36E-01			3.33E-01	8.10E-02	1.71E-01	1.70E-01	3.97E-01				3.33E-01	7.53E-02	2.16E-01	1.73E-01	4.16E-01		4.93E-01	2.06E-01	
18	3.67E-01	4.05E-02	3.01E-01	5.78E-02	5.78E-01			3.67E-01	9.04E-02	1.81E-01	2.83E-01	3.12E-01				3.67E-01	8.45E-02	2.36E-01	2.77E-01	5.89E-01		4.14E-01	1.80E-01	
19	4.00E-01	4.62E-02	3.18E-01	1.73E-01	5.20E-01			4.00E-01	9.80E-02	1.93E-01	2.75E-01	3.40E-01				4.00E-01	8.92E-02	2.49E-01	1.39E-01	3.81E-01		3.60E-01	1.43E-01	
20	4.33E-01	4.62E-02	3.33E-01	0.00E+00	4.62E-01			4.33E-01	1.06E-01	2.04E-01	2.55E-01	3.40E-01				4.33E-01	9.49E-02	2.58E-01	1.73E-01	2.77E-01		5.19E-01	1.61E-01	
21	4.67E-01	4.62E-02	3.54E-01	0.00E+00	6.36E-01			4.67E-01	1.12E-01	2.12E-01	1.70E-01	2.27E-01				4.67E-01	1.05E-01	2.81E-01	3.12E-01	6.93E-01		2.89E-01	1.98E-01	
22	5.00E-01	5.20E-02	3.66E-01	1.73E-01	3.47E-01			5.00E-01	1.22E-01	2.22E-01	2.83E-01	3.12E-01				5.00E-01	1.10E-01	2.88E-01	1.39E-01	2.08E-01		4.00E-01	1.05E-01	
23	5.33E-01	5.39E-02	3.85E-01	5.78E-02	5.78E-01			5.33E-01	1.24E-01	2.28E-01	8.50E-02	1.70E-01				5.33E-01	1.16E-01	3.03E-01	1.73E-01	4.51E-01		3.24E-01	1.95E-01	
24	5.67E-01	5.78E-02	3.99E-01	1.16E-01	4.05E-01			5.67E-01	1.32E-01	2.36E-01	2.27E-01	2.55E-01				5.67E-01	1.24E-01	3.13E-01	2.43E-01	3.12E-01		3.37E-01	1.45E-01	
25	6.00E-01	5.97E-02	4.10E-01	5.78E-02	3.47E-01			6.00E-01	1.38E-01	2.46E-01	1.70E-01	2.83E-01				6.00E-01	1.31E-01	3.26E-01	2.08E-01	3.81E-01		2.15E-01	1.09E-01	
26	6.33E-01	6.16E-02	4.20E-01	5.78E-02	2.89E-01			6.33E-01	1.44E-01	2.49E-01	1.98E-01	1.13E-01				6.33E-01	1.33E-01	3.34E-01	6.93E-02	2.43E-01		2.45E-01	1.60E-01	
27	6.67E-01	6.55E-02	4.28E-01	1.16E-01	2.31E-01			6.67E-01	1.52E-01	2.57E-01	2.27E-01	2.27E-01				6.67E-01	1.38E-01	3.43E-01	1.39E-01	2.77E-01		2.11E-01	1.57E-01	
28	7.00E-01	6.74E-02	4.39E-01	5.78E-02	3.47E-01			7.00E-01	1.58E-01	2.61E-01	1.70E-01	1.13E-01				7.00E-01	1.46E-01	3.49E-01	2.43E-01	1.73E-01		2.29E-01	2.15E-01	
29	7.33E-01	7.32E-02	4.47E-01	1.73E-01	2.31E-01			7.33E-01	1.70E-01	2.70E-01	3.68E-01	2.83E-01				7.33E-01	1.49E-01	3.55E-01	1.04E-01	1.73E-01		1.47E-01	1.19E-01	
30	7.67E-01	7.51E-02	4.57E-01	5.78E-02	2.89E-01			7.67E-01	1.72E-01	2.68E-01	5.67E-02	-5.67E-02				7.67E-01	1.57E-01	3.62E-01	2.43E-01	2.08E-01		1.76E-01	2.02E-01	
31	8.00E-01	7.71E-02	4.62E-01	5.78E-02	1.73E-01			8.00E-01	1.83E-01	2.72E-01	3.40E-01	1.13E-01				8.00E-01	1.64E-01	3.70E-01	2.08E-01	2.43E-01		1.70E-01	1.38E-01	
32	8.33E-01	8.48E-02	4.76E-01	2.31E-01	4.05E-01			8.33E-01	1.87E-01	2.72E-01	1.13E-01	0.00E+00				8.33E-01	1.67E-01	3.73E-01	6.93E-02	1.04E-01		1.23E-01	1.09E-01	
33	8.67E-01	8.28E-02	4.84E-01	5.78E-02	2.31E-01			8.67E-01	1.92E-01	2.72E-01	1.42E-01	0.00E+00				8.67E-01	1.75E-01	3.78E-01	2.43E-01	1.39E-01		5.60E-02	1.62E-01	
34	9.00E-01	8.67E-02	4.85E-01	1.16E-01	5.78E-02			9.00E-01	1.98E-01	2.71E-01	1.98E-01	-2.83E-02				9.00E-01	1.80E-01	3.83E-01	1.73E-01	1.39E-01		3.47E-02	1.14E-01	
35	9.33E-01	8.67E-02	4.85E-01	0.00E+00	0.00E+00			9.33E-01	2.04E-01	2.71E-01	1.70E-01	0.00E+00				9.33E-01	1.86E-01	3.86E-01	1.73E-01	1.04E-01		5.60E-02	1.33E-01	
36	9.67E-01	8.67E-02	4.87E-01	0.00E+00	5.78E-02			9.67E-01	2.11E-01	2.70E-01	2.27E-01	-2.83E-02				9.67E-01	1.92E-01	3.91E-01	1.73E-01	1.39E-01		3.82E-02	2.09E-01	
37	1.00E+00	9.63E-02	4.89E-01	2.89E-01	5.78E-02			1.00E+00	2.18E-01	2.72E-01	1.98E-01	5.67E-02				1.00E+00	1.97E-01	3.91E-01	1.39E-01	0.00E+00		1.35E-01	9.35E-02	
38	1.03E+00	9.63E-02	5.05E-01	0.00E+00	4.62E-01			1.03E+00	2.23E-01	2.70E-01	1.42E-01	-5.67E-02				1.03E+00	2.01E-01	3.91E-01	1.39E-01	0.00E+00		-1.99E-02	1.45E-01	
39	1.07E+00	9.63E-02	5.05E-01	0.00E+00	0.00E+00			1.07E+00	2.30E-01	2.63E-01	2.27E-01	-1.98E-01				1.07E+00	2.08E-01	3.95E-01	2.08E-01	1.39E-01		-1.26E-02	1.05E-01	
40	1.10E+00	9.63E-02	5.05E-01	0.00E+00	0.00E+00			1.10E+00	2.35E-01	2.59E-01	1.42E-01	-1.42E-01				1.10E+00	2.14E-01	3.99E-01	1.73E-01	1.04E-01		-3.78E-02	2.08E-01	
41	1.13E+00	1.04E-01	5.05E-01	2.31E-01	0.00E+00			1.13E+00	2.43E-01	2.55E-01	2.55E-01	-1.13E-01				1.13E+00	2.19E-01	3.99E-01	1.39E-01	0.00E+00		-9.35E-02	2.66E-01	
42	1.17E+00	1.04E-01	5.05E-01	0.00E+00	0.00E+00			1.17E+00	2.49E-01	2.50E-01	1.70E-01	-1.42E-01				1.17E+00	2.26E-01	3.94E-01	2.08E-01	-1.39E-01		-8.61E-02	7.00E-02	
43	1.20E+00	1.02E-01	4.99E-01	-5.78E-02	-1.73E-01			1.20E+00	2.56E-01	2.47E-01	1.98E-01	-8.50E-02				1.20E+00	2.28E-01	3.94E-01	6.93E-02	0.00E+00		-3.71E-02	2.43E-01	
44	1.23E+00	1.12E-01	5.03E-01	2.89E-01	1.16E-01			1.23E+00	2.62E-01	2.40E-01	1.98E-01	-2.27E-01				1.23E+00	2.36E-01	3.94E-01	6.93E-02	0.00E+00		-1.03E-01	1.01E-01	
45	1.27E+00	1.12E-01	5.03E-01	0.00E+00	0.00E+00			1.27E+00	2.69E-01	2.34E-01	1.98E-01	-1.70E-01				1.27E+00	2.39E-01	3.90E-01	1.04E-01	-1.39E-01		-1.61E-01	1.47E-01	
46	1.30E+00	1.14E-01	4.97E-01	5.78E-02	-1.73E-01			1.30E+00	2.74E-01	2.29E-01	1.42E-01	-1.70E-01				1.30E+00	2.47E-01	3.85E-01	2.43E-01	-1.39E-01		-4.86E-02	1.54E-01	
47	1.33E+00	1.14E-01	5.01E-01	0.00E+00	1.16E-01			1.33E+00	2.82E-01	2.21E-01	2.55E-01	-2.27E-01				1.33E+00	2.54E-01	3.84E-01	2.08E-01	-3.47E-02		-1.60E-01	1.33E-01	
48	1.37E+00	1.14E-01	4.97E-01	0.00E+00	-1.16E-01			1.37E+00	2.90E-01	2.13E-01	2.27E-01	-2.27E-01				1.37E+00	2.60E-01	3.79E-01	1.73E-01	-1.39E-01		-1.44E-01	1.09E-01	
49	1.40E+00	1.16E-01	4.93E-01	5.78E-02	-1.16E-01			1.40E+00	2.96E-01	2.09E-01	1.98E-01	-1.42E-01				1.40E+00	2.62E-01	3.73E-01	6.93E-02	-1.73E-01		-2.03E-01	1.91E-01	
50	1.43E+00	1.21E-01	4.89E-01	1.73E-01	-1.16E-01			1.43E+00	3.04E-01	1.95E-01	2.27E-01	-4.25E-01				1.43E+00	2.68E-01	3.71E-01	1.73E-01	-6.93E-02		-1.62E-01	1.07E-01	
51	1.47E+00	1.21E-01	4.85E-01	0.00E+00	-1.16E-01			1.47E+00	3.08E-01	1.88E-01	1.13E-01	-1.98E-01												